

Q1 $f(x) = -\sqrt{\frac{x}{x-1}}$

denom $\neq 0$

even root denom > 0

$(1, \infty)$

$x \neq 1$

$\frac{2}{2-1} = 1.4$ will work

$x = 0$

$\frac{2}{0-1}$ error won't work

Q3 $\lim_{t \rightarrow 0} \left(\frac{1}{t^2+6} + \frac{1}{t} \right)$

0.01

$\frac{1}{(0.01)^2+0.01} - \frac{1}{0.01}$

-0.990099

$t(t+1)$

0.001

$\frac{1}{(0.001)^2+0.001} - \frac{1}{0.001}$

-0.999001

-0.01

1

0.0001

$\frac{1}{(-0.01)^2+0.01} - \frac{1}{(-0.01)}$

$\frac{1}{(0.0001)^2+0.0001} - \frac{1}{0.0001}$

-0.999900

-1.010101

-0.001

1

1

$\frac{1}{(-0.001)^2-0.001} - \frac{1}{(-0.001)}$

-1.001001

Q4 $\lim_{x \rightarrow 2\pi^-} x \csc(x)$

$$\frac{1}{\sin x} \quad \frac{2\pi}{0^-} = -\infty$$

As $x \rightarrow 2\pi^-$

$x \rightarrow 2\pi$ positive

$\sin(x) \rightarrow 0$

$(\pi, 2\pi), \sin x < 0$

$x \rightarrow 2\pi^-$

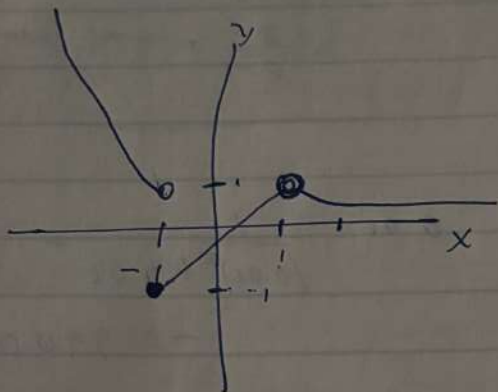
$\sin(x) \approx 0^-$

$x^2 \quad -1^2 = 1$

$x \quad -1 = -1$

$\frac{1}{x} \quad \frac{1}{-1} = -1$

Q5



Q6 $\frac{(x-3)(x+3)}{(x-3)(x+5)}$

$\frac{3+3}{3+5} = \frac{6}{8} = \frac{3}{4}$

Q7 $\frac{\sqrt{x} + x}{x^2}$

$\frac{x^{1/2} + x}{x^2}$

$\frac{-3/2}{x} + \frac{-1}{x}$

$f' \quad \frac{1}{2}x^{-1/2} + 1$

$\left[-\frac{3}{2}x^{-5/2} - \frac{-2}{x} \right]$

$-\frac{1}{2}x^{-3/2} + (-1)x^{-2}$

$-\frac{1}{2x^{3/2}} - \frac{1}{x^2}$

$g' = 2x$

2x

$$\sin(x+y) = z_x - z_y$$

$$v = x + y$$

$$y = \sin v$$

$$\sin x + \sin y = z_x - z_y$$

$$z_y + \sin y = z_x - \sin x$$

$$y (z + \sin)$$

$$y = \frac{z_x - \sin x}{z + \sin}$$

$$\frac{z'' - 0}{z + 0} = \frac{0}{11}$$

$$\frac{z - \cos(11)}{z + 11} = \frac{0}{11}$$

$$x + y = 211$$

$$\cos(211) = 1$$

$$y - 11 = \frac{1}{3}(x - 11)$$

$$1(1+y') = z - z_y'$$

$$1 + y' = z - z_y$$

$$3y' = 1$$

$$y = \frac{1}{3}$$

$$m = \frac{1}{3}$$

Q11 $P(t) = P_0 e^{kt}$ $2050 - 2022 = 28$

$$P_0 = 3.25$$

$$P(28) = 1.82$$

$$1.82 = 3.25 e^{28k}$$

$$\frac{1.82}{3.25} = e^{28k}$$

$$k = \frac{1}{28} \ln \left(\frac{1.82}{3.25} \right)$$

$$f = \cos x \quad f' = -\sin x$$

$$g = 1 - \sin x \quad g' = -\cos x$$

$$\frac{(1 - \sin x) - \sin x}{(\cos x)(-\cos x)} = \frac{-\sin x + \sin^2 x}{-\cos^2 x}$$

$$\frac{-\sin x + \sin^2 x - (-\cos^2 x)}{(1 - \sin x)^2}$$

$$\frac{-\sin x + 1}{(1 - \sin x)}$$

$$\frac{1 - \sin x}{1}$$

Q2 $x < 0$ num + denom $\rightarrow -x$
 $0 < x < 1$ num + denom $\rightarrow -x$
 $x > 1$ num + denom $\rightarrow + \checkmark$
 $x = 0$ $0 \rightarrow \checkmark$

$$x = 0$$

$$f(0) = -\sqrt{0} = 0$$

$$\text{As } x \rightarrow 1^+ \quad \frac{x^2}{x-1} \rightarrow \infty$$

$$(-\infty, 0]$$